Instituto Oficial de Formación Profesional

Servidor VoiP
Asterisk

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Índice

Servidor VoiP Asterisk

1. Instalación de Asterisk

Para poder instalar el servicio de Asterisk tendremos que actualizar el sistema con el **comando sudo apt update y sudo apt upgrade -y**

Cuando ya lo tengamos actualizado ejecutaremos el **comando wget** **https://downloads.asterisk.org/pub/telephony/asterisk/asterisk-21-current.tar.gz**

Creamos una carpeta llamada asterisk y otra llamada src con el **comando mkdir (nombre de la carpeta)**

Cuando tengamos el archivo de instalación lo moveremos a la carpeta de src con el **comando sudo mv asterisk-21-current.tar.gz /asterisk/src**

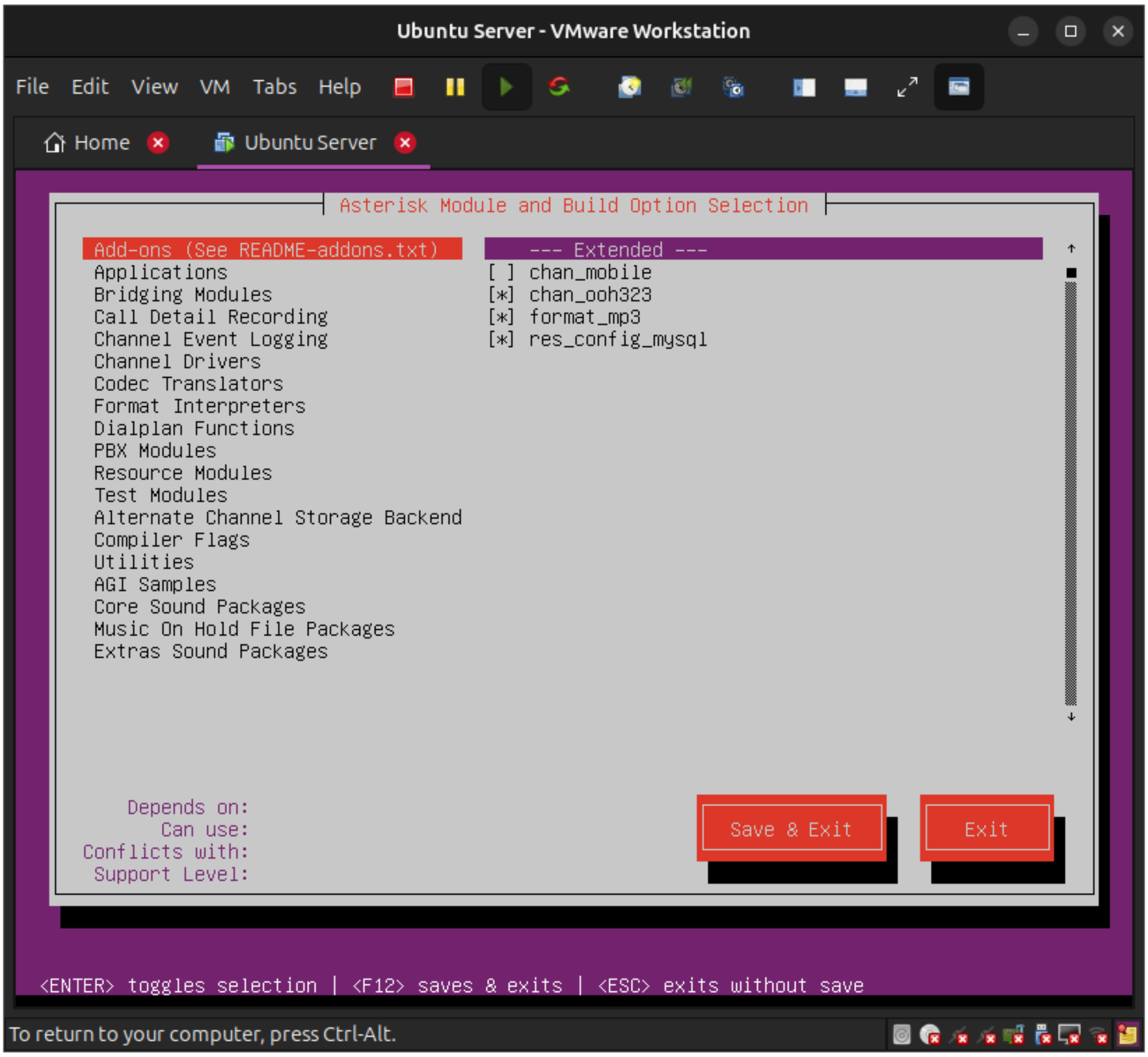
Ejecutaremos el **comando sudo tar -xvzf asterisk-21-current.tar.gz**

Nos meteremos a la carpeta con el **comando cd asterisk-21-current** y ejecutaremos el **comando contrib/scripts/get_mp3_source.sh** para ejecutar ese archivo

Ahora instalaremos las dependencias con el **comando sudo contrib/scripts/install_prereq install**

Configuraremos asterisk con el **comando sudo ./configure**

Ahora instalaremos los siguientes módulos con el **comando sudo make menuselect**


Ubuntu Server - VMware Workstation
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Ubuntu Server
Asterisk Module and Build Option Selection
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Dialplan Functions
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Test Modules
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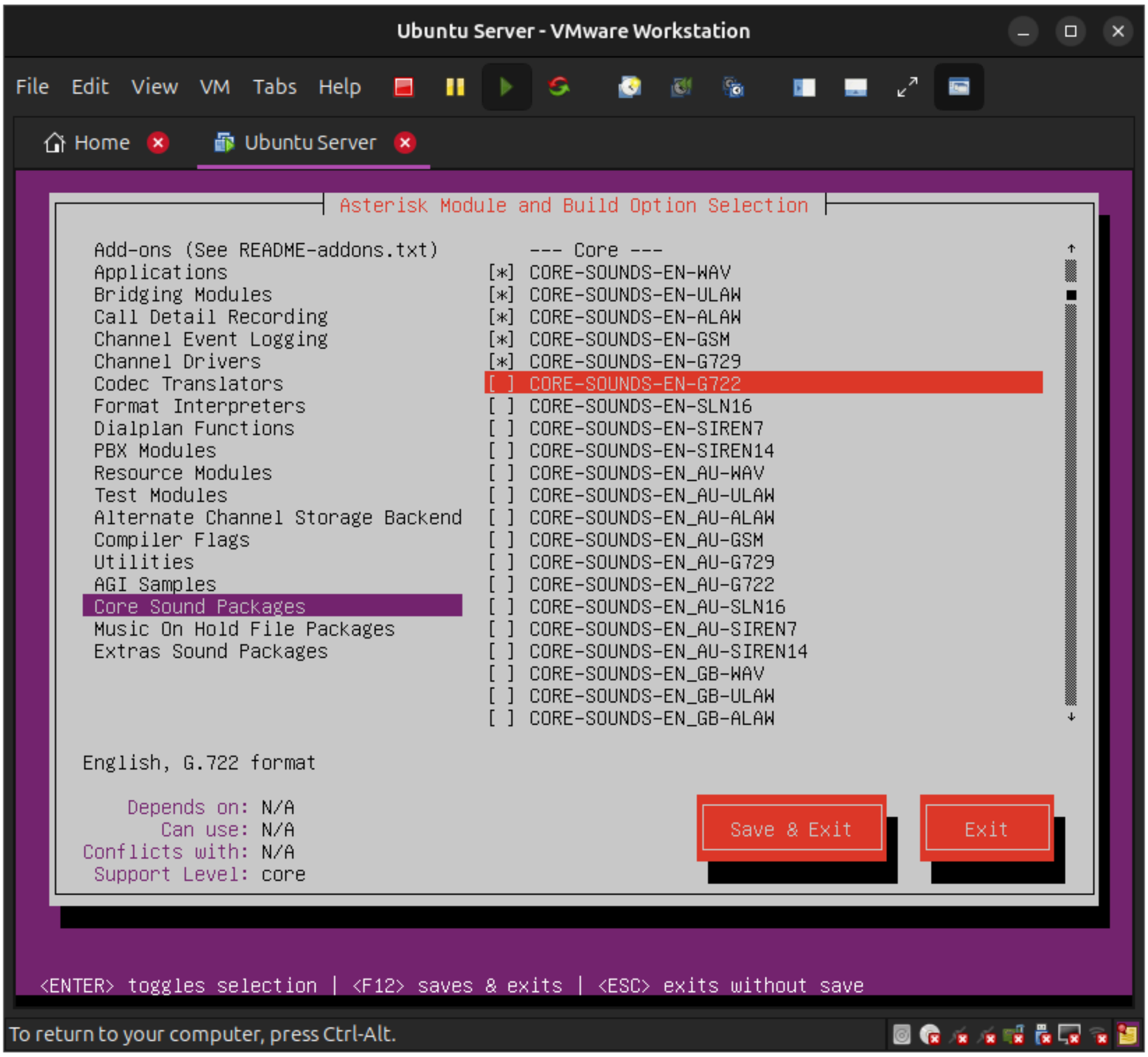
AGI Samples
Core Sound Packages
Music On Hold File Packages
Extras Sound Packages
--- Extended ---
[ ] chan_mobile
[*] chan_ooh323
[*] format_mp3
[*] res_config_mysql
Depends on:
Can use:
Conflicts with:
Support Level:
Save & Exit
Exit
<ENTER> toggles selection | <F12> saves & exits | <ESC> exits without save
To return to your computer, press Ctrl-Alt.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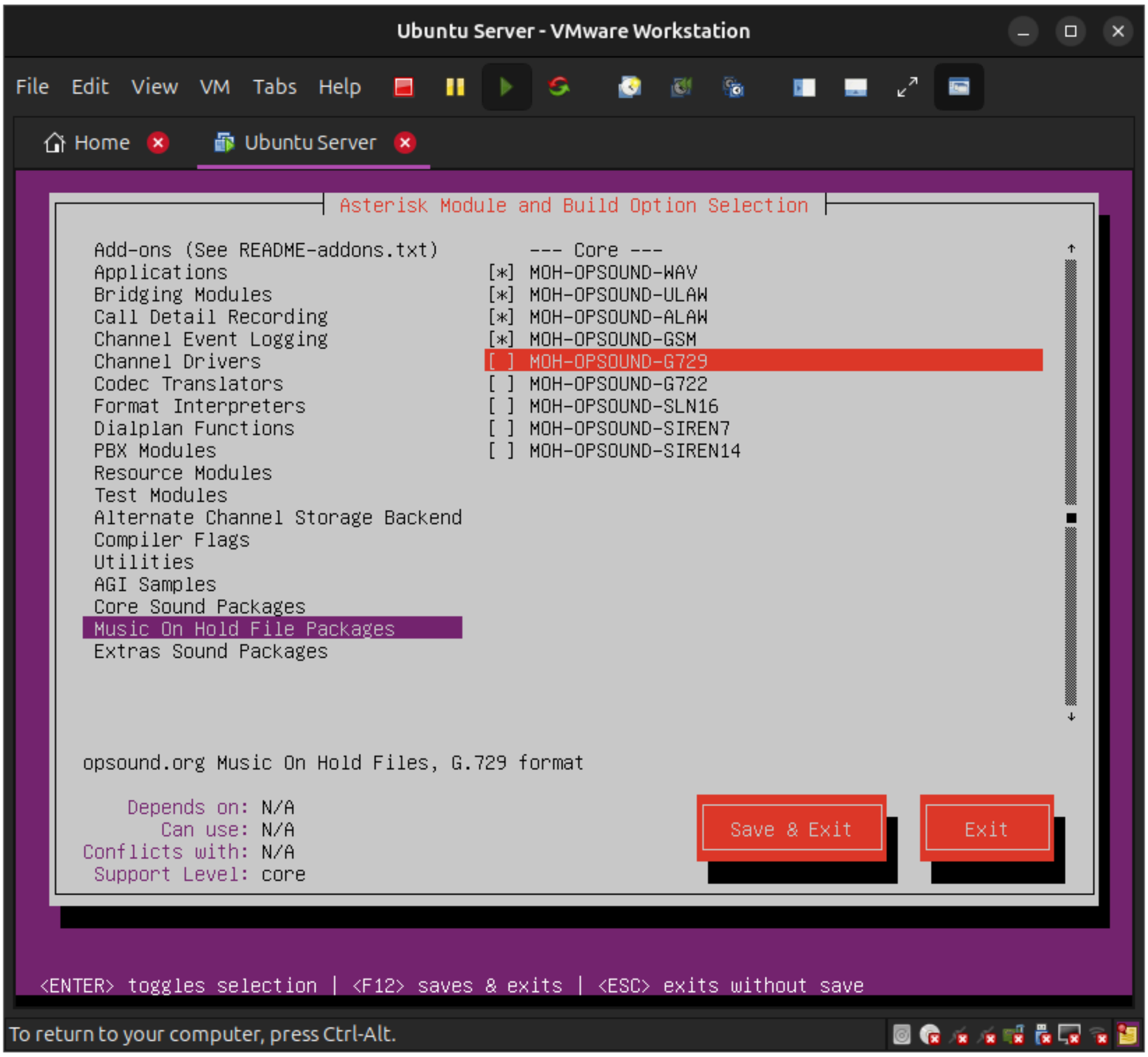
--- Core ---
[*] CORE-SOUNDS-EN-WAV
[*] CORE-SOUNDS-EN-ULAW
[*] CORE-SOUNDS-EN-ALAW
[*] CORE-SOUNDS-EN-GSM
[*] CORE-SOUNDS-EN-G729
[ ] CORE-SOUNDS-EN-G722
[ ] CORE-SOUNDS-EN-SLN16
[ ] CORE-SOUNDS-EN-SIREN7
[ ] CORE-SOUNDS-EN-SIREN14
[ ] CORE-SOUNDS-EN_AU-WAV
[ ] CORE-SOUNDS-EN_AU-ULAW
[ ] CORE-SOUNDS-EN_AU-ALAW
[ ] CORE-SOUNDS-EN_AU-GSM
[ ] CORE-SOUNDS-EN_AU-G729
[ ] CORE-SOUNDS-EN_AU-G722
[ ] CORE-SOUNDS-EN_AU-SLN16
[ ] CORE-SOUNDS-EN_AU-SIREN7
[ ] CORE-SOUNDS-EN_AU-SIREN14
[ ] CORE-SOUNDS-EN_GB-WAV
[ ] CORE-SOUNDS-EN_GB-ULAW
[ ] CORE-SOUNDS-EN_GB-ALAW
English, G.722 format
Depends on: N/A
Can use: N/A
Conflicts with: N/A
Support Level: core
Save & Exit
Exit
<ENTER> toggles selection | <F12> saves & exits | <ESC> exits without save
To return to your computer, press Ctrl-Alt.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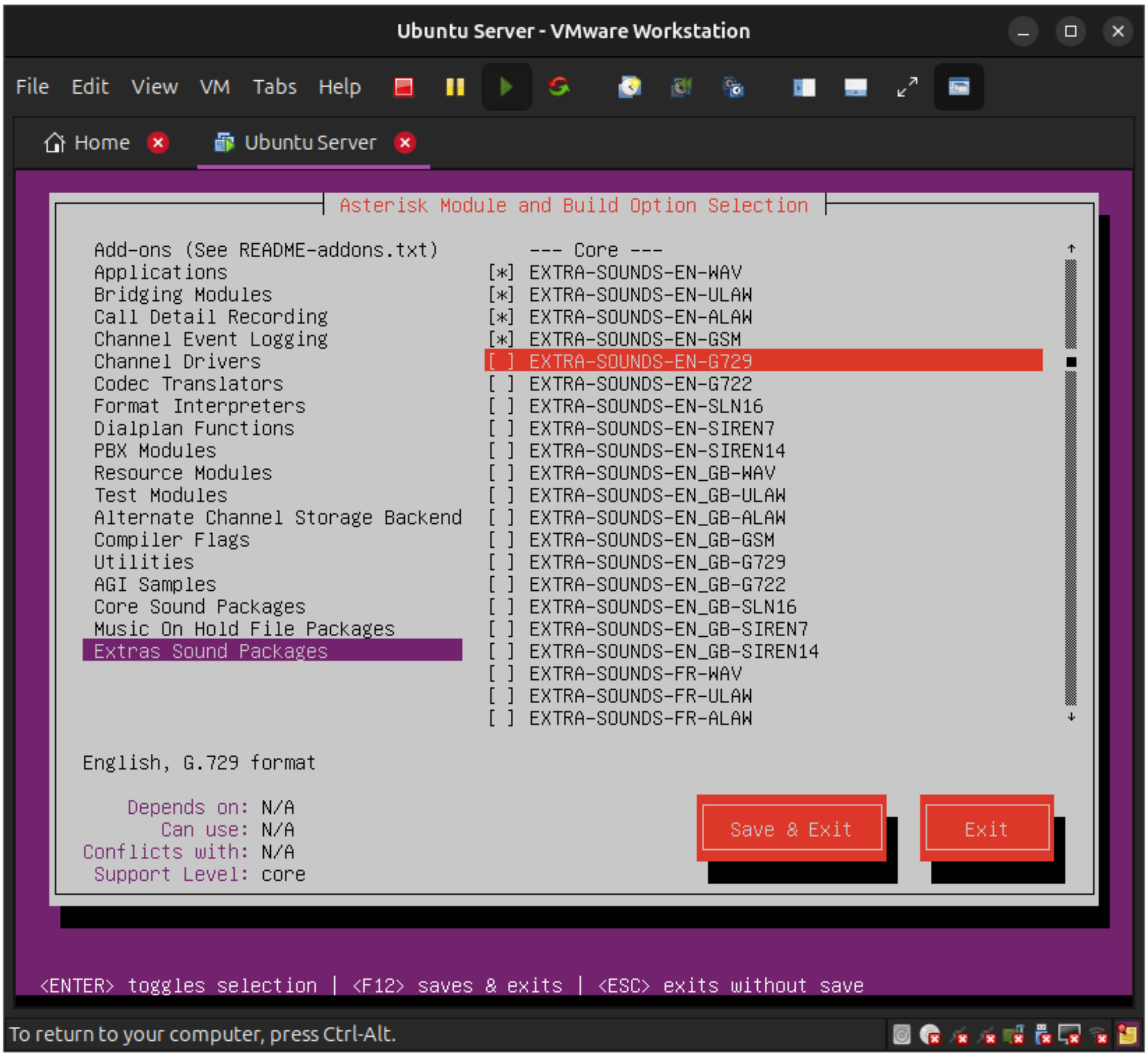
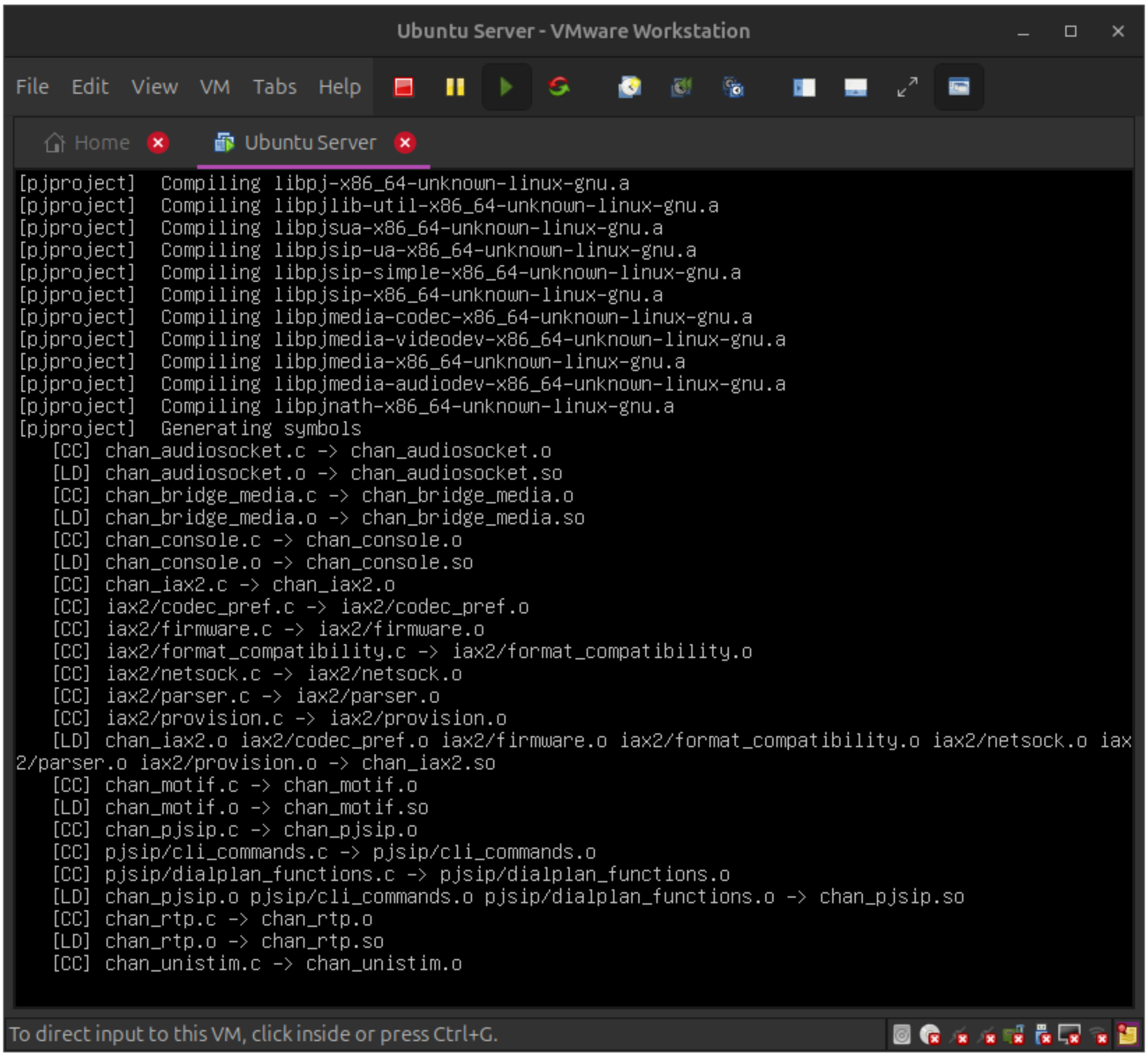
[*] MOH-OPSOUND-WAV
[*] MOH-OPSOUND-ULAW
[*] MOH-OPSOUND-ALAW
[*] MOH-OPSOUND-GSM
[ ] MOH-OPSOUND-G729
[ ] MOH-OPSOUND-G722
[ ] MOH-OPSOUND-SLN16
[ ] MOH-OPSOUND-SIREN7
[ ] MOH-OPSOUND-SIREN14
opsound.org Music On Hold Files, G.729 format
Depends on: N/A
Can use: N/A
Conflicts with: N/A
Support Level: core
Save & Exit
Exit
<ENTER> toggles selection | <F12> saves & exits | <ESC> exits without save
To return to your computer, press Ctrl-Alt.



Ahora haremos la construcción de asterisk con el **comando sudo make**, ejecutaremos el c**omando sudo make install** para la instalación, el **comando sudo make samples y sudo make config** y ejecutaremos el **comando ldconfig** para ver que se haya creado correctamente


Ubuntu Server - VMware Workstation
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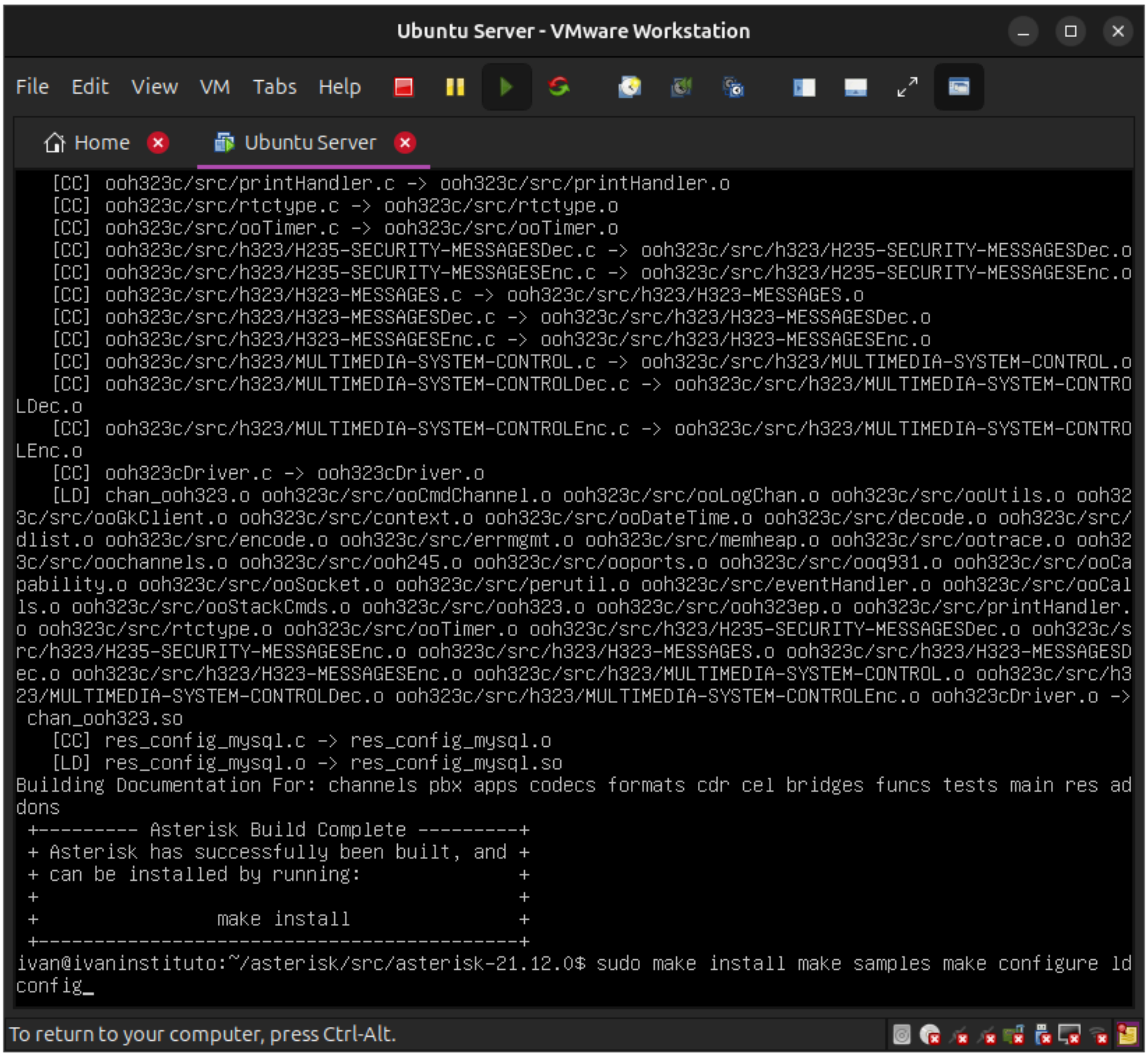
Ubuntu Server
[pjproject] Compiling libpj-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjlib-util-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjsua-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjsip-ua-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjsip-simple-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjsip-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjmedia-codec-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjmedia-videodev-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjmedia-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjmedia-audiodev-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjnath-x86_64-unknown-linux-gnu.a
[pjproject] Generating symbols
[CC] chan_audiosocket.c -> chan_audiosocket.o
[LD] chan_audiosocket.o -> chan_audiosocket.so
[CC] chan_bridge_media.c -> chan_bridge_media.o
[LD] chan_bridge_media.o -> chan_bridge_media.so
[CC] chan_console.c -> chan_console.o
[LD] chan_console.o -> chan_console.so
[CC] chan_iax2.c -> chan_iax2.o
[CC] iax2/codec_pref.c -> iax2/codec_pref.o
[CC] iax2/firmware.c -> iax2/firmware.o
[CC] iax2/format_compatibility.c -> iax2/format_compatibility.o
[CC] iax2/netsock.c -> iax2/netsock.o
[CC] iax2/parser.c -> iax2/parser.o
[CC] iax2/provision.c -> iax2/provision.o
[LD] chan_iax2.o iax2/codec_pref.o iax2/firmware.o iax2/format_compatibility.o iax2/netsock.o iax2/parser.o iax2/provision.o -> chan_iax2.so
[CC] chan_motif.c -> chan_motif.o
[LD] chan_motif.o -> chan_motif.so
[CC] chan_pjsip.c -> chan_pjsip.o
[CC] pjsip/cli_commands.c -> pjsip/cli_commands.o
[CC] pjsip/dialplan_functions.c -> pjsip/dialplan_functions.o
[LD] chan_pjsip.o pjsip/cli_commands.o pjsip/dialplan_functions.o -> chan_pjsip.so
[CC] chan_rtp.c -> chan_rtp.o
[LD] chan_rtp.o -> chan_rtp.so
[CC] chan_unistim.c -> chan_unistim.o
To direct input to this VM, click inside or press Ctrl+G.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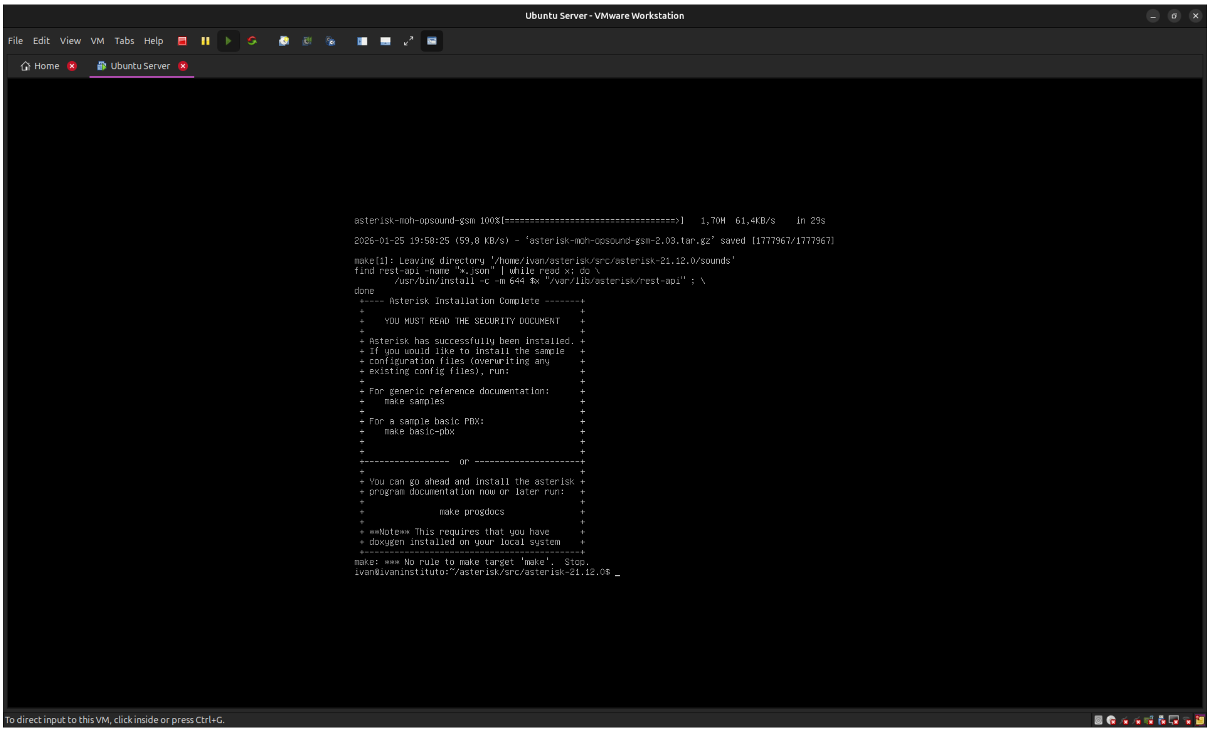
[CC] ooh323c/src/printHandler.c -> ooh323c/src/printHandler.o
[CC] ooh323c/src/rtctype.c -> ooh323c/src/rtctype.o
[CC] ooh323c/src/ooTimer.c -> ooh323c/src/ooTimer.o
[CC] ooh323c/src/h323/H235-SECURITY-MESSAGESDec.c -> ooh323c/src/h323/H235-SECURITY-MESSAGESDec.o
[CC] ooh323c/src/h323/H235-SECURITY-MESSAGESEnc.c -> ooh323c/src/h323/H235-SECURITY-MESSAGESEnc.o
[CC] ooh323c/src/h323/H323-MESSAGES.c -> ooh323c/src/h323/H323-MESSAGES.o
[CC] ooh323c/src/h323/H323-MESSAGESDec.c -> ooh323c/src/h323/H323-MESSAGESDec.o
[CC] ooh323c/src/h323/H323-MESSAGESEnc.c -> ooh323c/src/h323/H323-MESSAGESEnc.o
[CC] ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROL.c -> ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROL.o
[CC] ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROLDec.c -> ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROLDec.o
[CC] ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROLEnc.c -> ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROLEnc.o
[CC] ooh323cDriver.c -> ooh323cDriver.o
[LD] chan_ooh323.o ooh323c/src/ooCmdChannel.o ooh323c/src/ooLogChan.o ooh323c/src/ooUtils.o ooh323c/src/ooGkClient.o ooh323c/src/context.o ooh323c/src/ooDateTime.o ooh323c/src/decode.o ooh323c/src/dlist.o ooh323c/src/encode.o ooh323c/src/errmgmt.o ooh323c/src/memheap.o ooh323c/src/ootrace.o ooh323c/src/oochannels.o ooh323c/src/ooh245.o ooh323c/src/ooports.o ooh323c/src/ooq931.o ooh323c/src/ooCapability.o ooh323c/src/ooSocket.o ooh323c/src/perutil.o ooh323c/src/eventHandler.o ooh323c/src/ooCalls.o ooh323c/src/ooStackCmds.o ooh323c/src/ooh323.o ooh323c/src/ooh323ep.o ooh323c/src/printHandler.o ooh323c/src/rtctype.o ooh323c/src/ooTimer.o ooh323c/src/h323/H235-SECURITY-MESSAGESDec.o ooh323c/src/h323/H235-SECURITY-MESSAGESEnc.o ooh323c/src/h323/H323-MESSAGES.o ooh323c/src/h323/H323-MESSAGESDec.o ooh323c/src/h323/H323-MESSAGESEnc.o ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROL.o ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROLDec.o ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROLEnc.o ooh323cDriver.o -> chan_ooh323.so
[CC] res_config_mysql.c -> res_config_mysql.o
[LD] res_config_mysql.o -> res_config_mysql.so
Building Documentation For: channels pbx apps codecs formats cdr cel bridges funcs tests main res addons
+--------- Asterisk Build Complete ---------+
+ Asterisk has successfully been built, and +
+ can be installed by running: +
+ +
+ make install +
+---+
ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$ sudo make install make samples make configure ldconfig_
To return to your computer, press Ctrl-Alt.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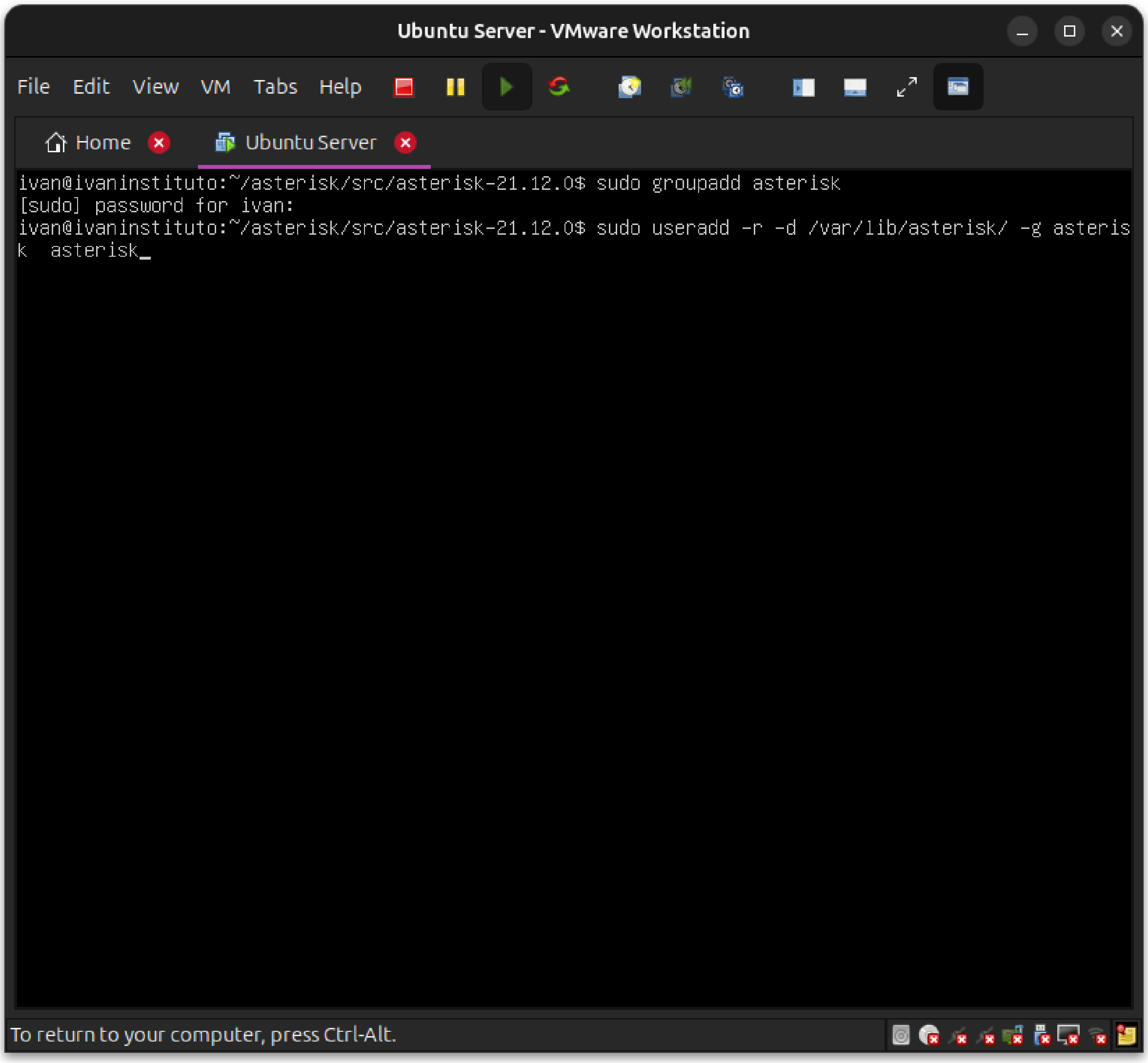
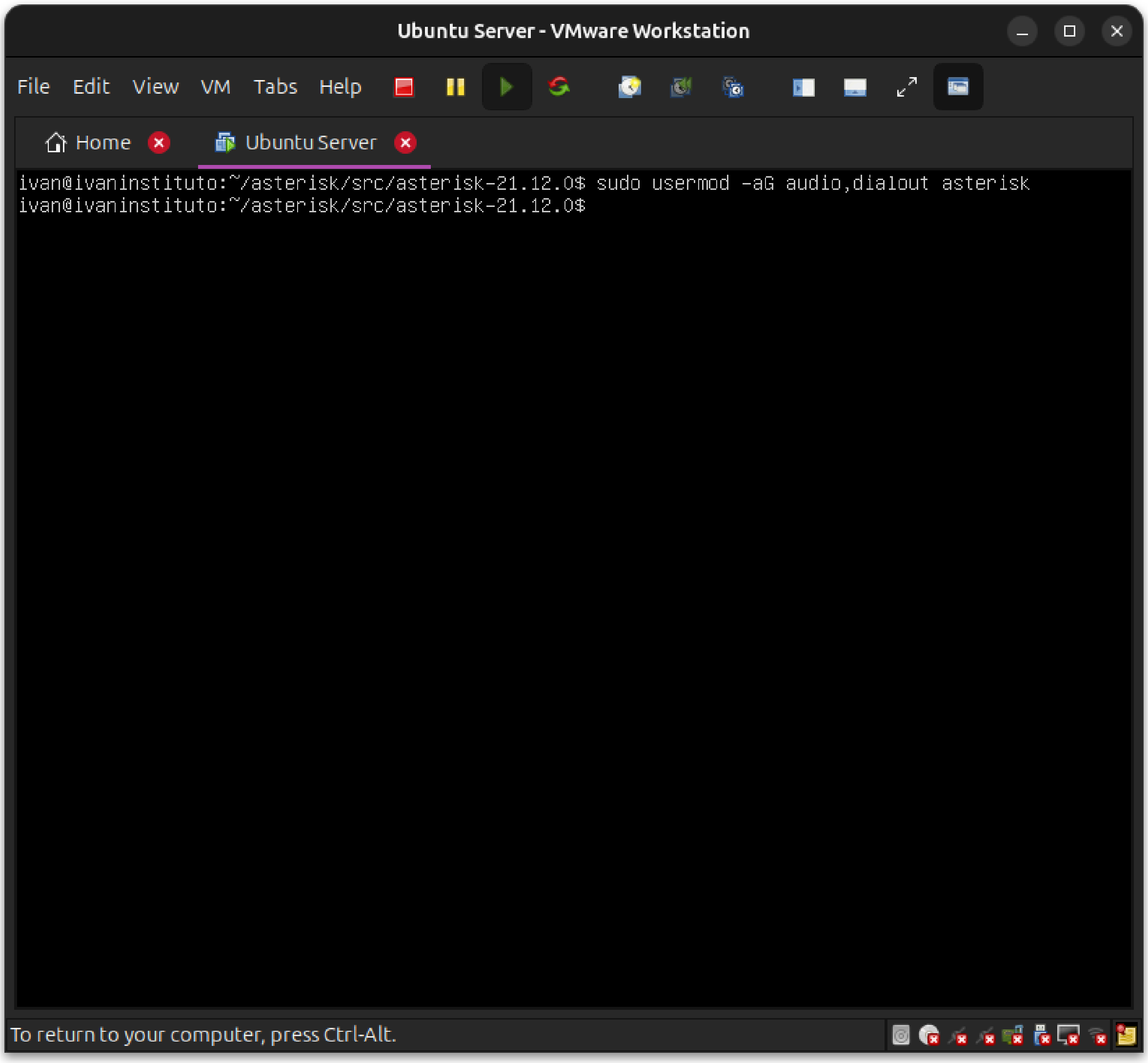
asterisk-moh-opsound-gsm 100%[===================================>] 1,70M 61,4KB/s in 29s
2026-01-25 19:58:25 (59,8 KB/s) - 'asterisk-moh-opsound-gsm-2.03.tar.gz' saved [1777967/1777967]
make[1]: Leaving directory '/home/ivan/asterisk/src/asterisk-21.12.0/sounds'
find rest-api -name "*.json" | while read x; do \
/usr/bin/install -c -m 644 $x "/var/lib/asterisk/rest-api" ; \
done
+---- Asterisk Installation Complete -------+
YOU MUST READ THE SECURITY DOCUMENT
Asterisk has successfully been installed.
If you would like to install the sample
configuration files (overwriting any
existing config files), run:
For generic reference documentation:
make samples
For a sample basic PBX:
make basic-pbx
or
You can go ahead and install the asterisk
program documentation now or later run:
make progdocs
Note This requires that you have
doxygen installed on your local system
make: *** No rule to make target 'make'. Stop.
ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$ _
To direct input to this VM, click inside or press Ctrl+G.

2. Creación de usuarios





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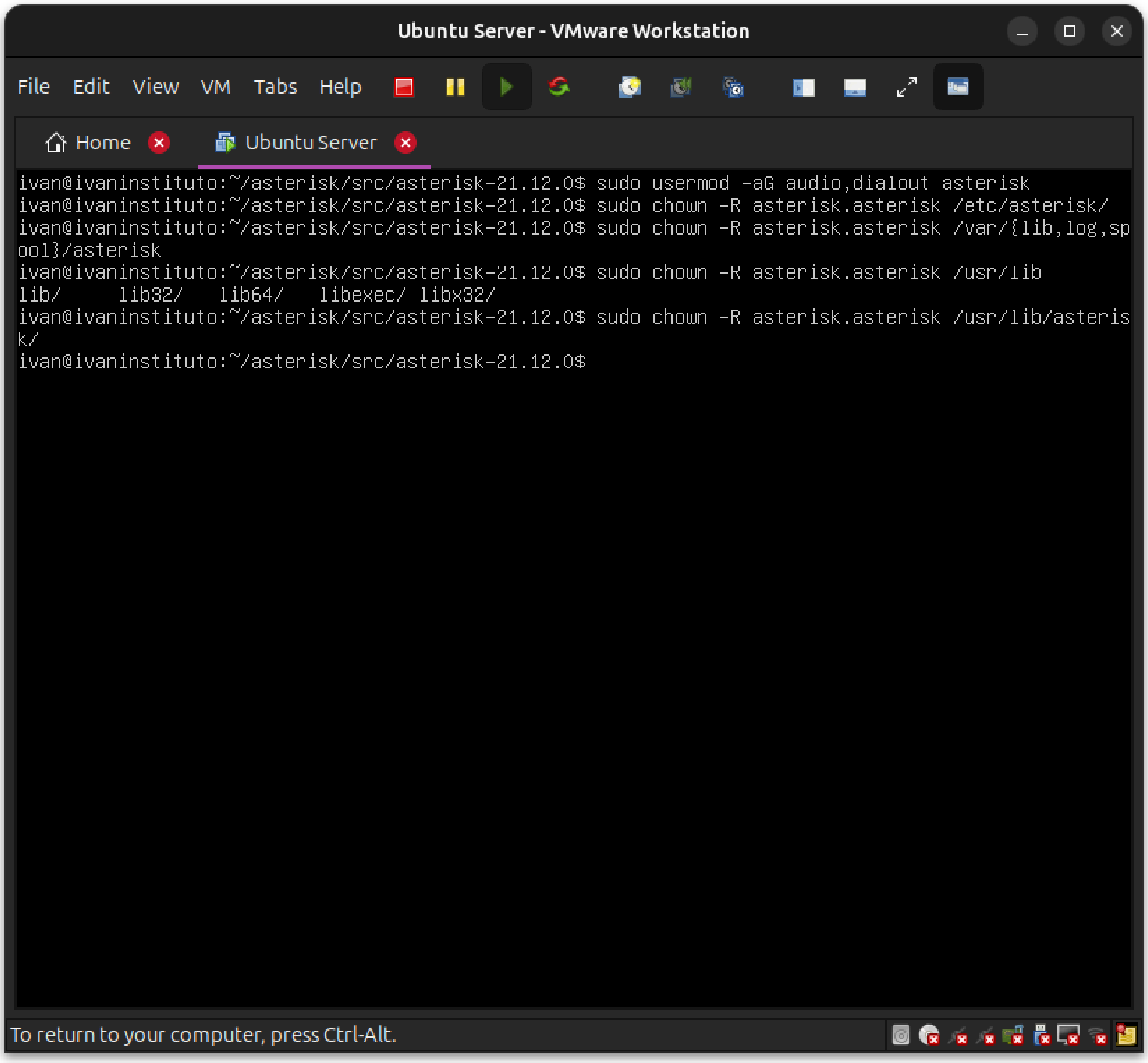
ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$ sudo usermod -aG audio,dialout asterisk
ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$
To return to your computer, press Ctrl-Alt.



Ahora crearemos los usuarios para asterisk con el comando **sudo groupadd asterisk** para añadir el **grupo** asterisk. El comando **sudo useradd -r -d /var/lib/asterisk -g asterisk asterisk** para crear el **usuario** de asterisk

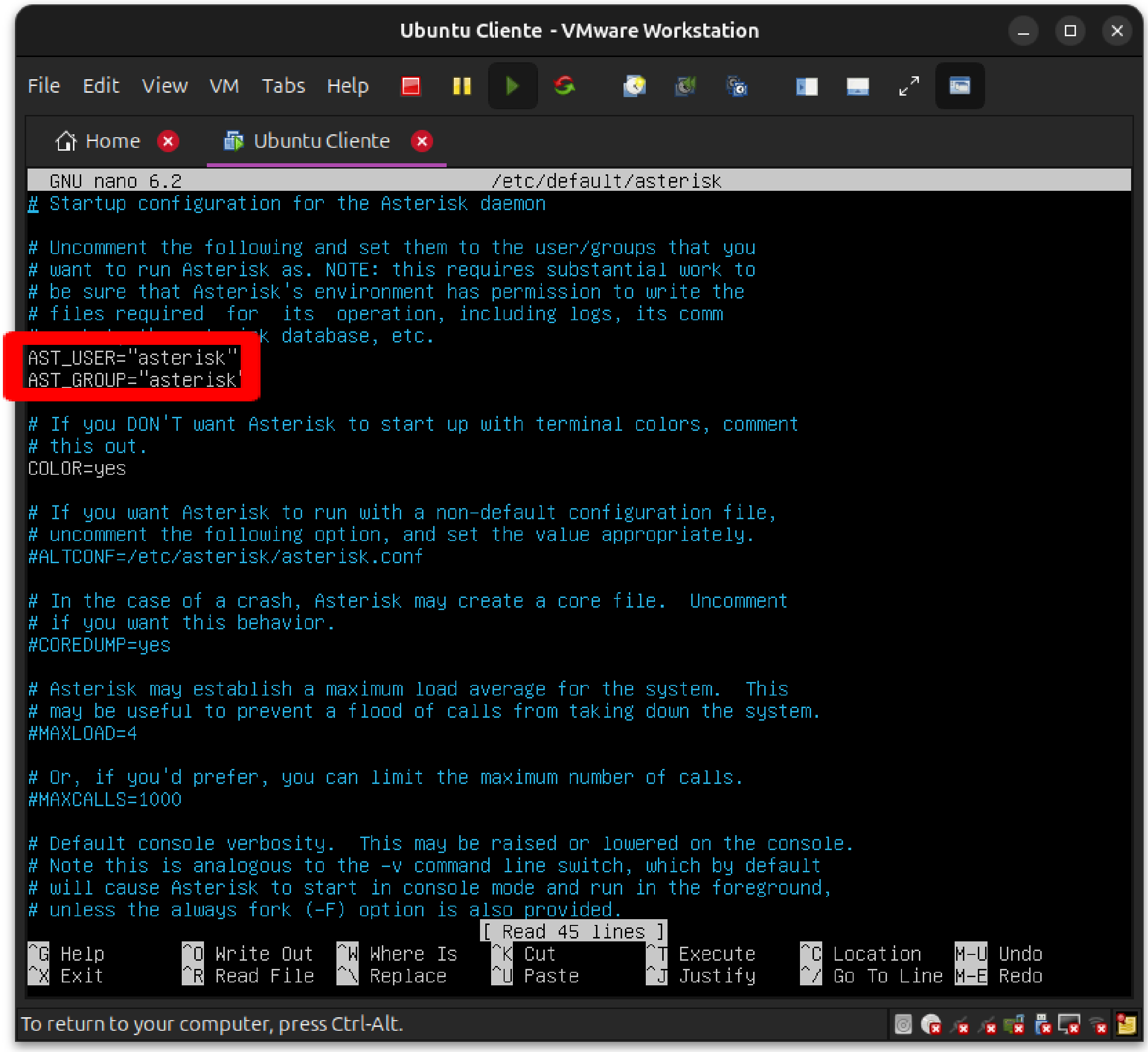
Añadiremos el usuario de audio y dialout al grupo de asterisk con el **comando sudo usermod -aG audio,dialout asterisk**

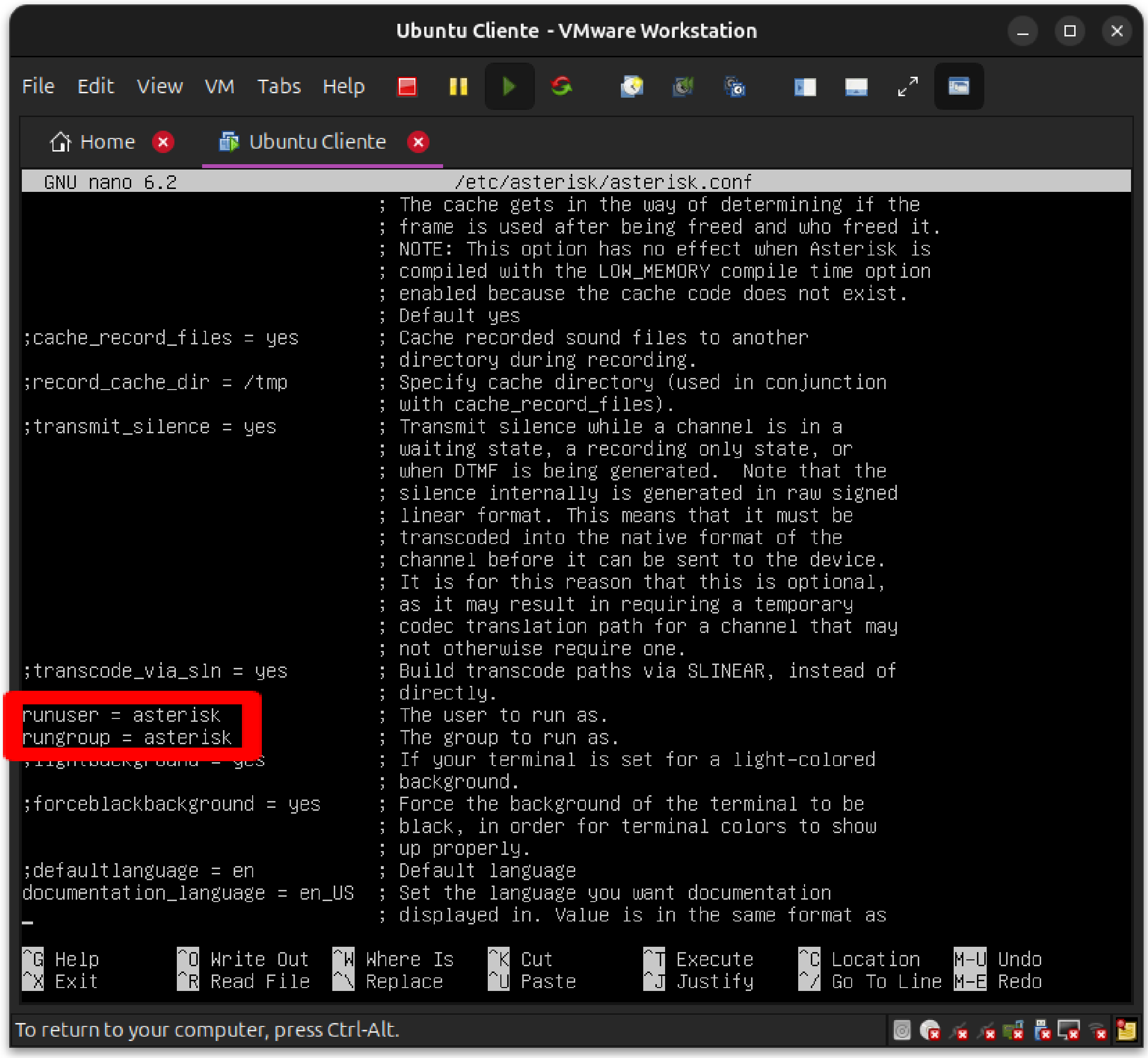
Cambiaremos la propiedad del directorio de configuración de asterisk con el **comando sudo chown -R asterisk.asterisk /etc/asterisk**

sudo chown -R asterisk.asterisk /var/{lib,log,spool}/asterisk

sudo chown -R asterisk.asterisk /usr/lib/asterisk

3. Configuración Asterisk





Ahora editaremos el archivo /etc/default/asterisk para establecer el usuario de asterisk por defecto

Descomentamos las siguientes líneas:

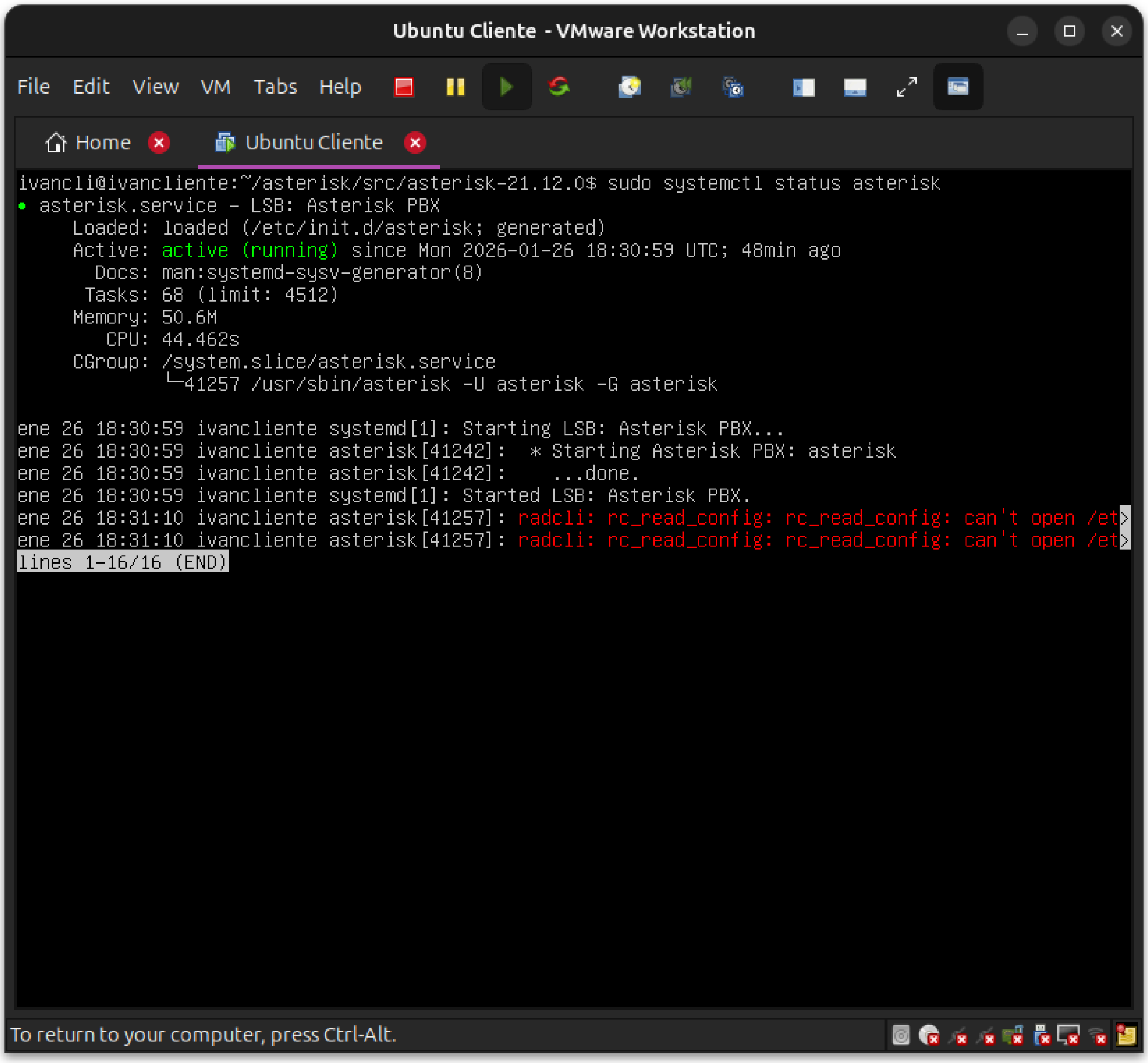
AST_USER=”asterisk”

AST_GROUP=”asterisk”

Editaremos el archivo /etc/asterisk/asterisk.conf y **descomentamos** las siguientes líneas:

runuser = asterisk ;The user to run as.

rungroup = asterisk ;The group to run as.



Por último ejecutaremos el **comando sudo systemctl restart asterisk** para que coja las configuraciones correctamente y lo activaremos con el **comando sudo systemctl enable asterisk**

Para terminar ejecutaremos el **comando sudo systemctl status asterisk** para comprobar que funciona correctamente, nos tiene que salir **active (running)** en color verde